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USB INTERFACE Using Python Software

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This is a simple transistor's output characteristic curve-tracer program through the interfacing of USB and PIC microcontroller using Python programming language. Output characteristic curve of a transistor is automated using Python software and USB interface. The application software for collecting the measured value and displaying it on the monitor screen is developed using Python.

Types of USB

In recent PCs, legacy ports such as parallel port, serial port and so on have been replaced with USB. A USB's plug-and-play feature can be used to automate laboratory instruments/data-acquisition systems. To meet the needs of various applications using USB interface, three speeds of operation have been designed in USB V2.0 specification, namely, low speed (1.5Mbps), full speed (12Mbps) and high speed (480Mbps).

Recently, USB3.0 specification has introduced super speed with 5Gbps. The physical USB connection uses four wires. Two wires are used to carry the differential signal (D+ and D-), and the other two are for power and ground. To transfer data between the host PC and the device USB V2.0 supports four types of data transfers, namely, control, bulk, interrupt and isochronous.

Control transfer is mainly used to configure a device when it is first connected to the PC. The process of configuration is known as enumeration. It is defined as the initial exchange of information by which the host learns about the device and assigns an appropriate device driver.

Bulk data transfers are used when data is generated or consumed in relatively large quantities.

PARTS LIST

Semiconductors:	
IC1	- PIC18F4550 microcontroller
IC2	- AD780 voltage regulator
IC3	- MAX5154 low-power, 12-bit digital-to-analogue converter
IC4	- LM358 low-power dual op-amp
LED1-LED2	- 5mm LED
D1-D3	- 1N4148 signal diode
T1	- BC547 npn transistor
T2	- SL100 npn power transistor
Resistors (all 1/4-watt, ±5% carbon):	
R1, R21	- 5.7-kilo-ohm
R2-R3	- 220-ohm
R4-R10	- 150-kilo-ohm
R11, R12	- 10-kilo-ohm
R13-R16	- 3.9-kilo-ohm
R17	- 47-kilo-ohm
R18	- 1-kilo-ohm
R19, R20	- 10-kilo-ohm
R22	- 1-ohm
Capacitors:	
C1	- 470pF ceramic disk
C2, C3	- 33pF ceramic disk
Miscellaneous:	
XTAL1	- 20MHz crystal oscillator
CON1	- 4-pin connector for USB A-type
CON2	- 2-pin connector for 12V battery
CON3	- 2-pin connector for 5V

Interrupt data transfers are used for timely but reliable delivery of data.

Isochronous data transfers occupy a pre-negotiated amount of USB bandwidth with pre-negotiated delivery latency (also called streaming real-time transfers).

One of the major barriers to designers of USB peripherals is developing device drivers for custom-built USB devices. This has been removed with the use of LibUSB-Win32, which is an open source driver.

Host software

USB supports master-slave configuration. Master is always the PC. Software that resides and controls the communication flow in the PC is known as host software. In this article, the host software used to communicate with the USB mass storage class/communication device class is Python.

Python is an interpreted language, which can save considerable time during program development because no compilation and linking are necessary. It is a popular programming language used for both standalone programs and scripting applications. It is free, portable, powerful and remarkably easy to use.

The interpreter can be used interactively, which makes it easy to experiment with features of the language. Programs written in Python are typically much shorter than equivalent C or C++ programs. There are two ways to use the interpreters, namely, interactive mode and script mode. In interactive mode, the interpreter prints the result as the program is typed.

The chevron, >>>, is the prompt the interpreter uses to indicate that it is ready. Alternatively, you can store the code in a file and use the interpreter to execute the contents of the file, which is called a script.

Software required. The required software include Python 2.5 (python-2.5.2) or higher IDE, Win32 Python (pywin32-210.win32-py2.5), Matplot Library (matplotlib-0.91.2.win32-Py2.5), Numeric Python (numpy-1.0.4.win32-py2.5) and PyUSB-1.0.0a2.

PyUSB-1.0.0a2 is a Python library allowing easy USB access. PyUSB-1.0.0a2 version is written in Python, which allows Python programmers with no background in C to understand how PyUSB works. PyUSB can run on any platform with Python 2.4 and later version. Communicating with a USB device has never been so easy. USB is a complex protocol, but PyUSB has all the necessary functions needed to configure a USB-supported device. PyUSB modules have two